

ECON120B: Final Exam Review

- OLS and estimators, and Causal treatment effect (before midterm)
- Multiple regression Model
 - Multicollinearity and Joint hypothesis testing
- Threats to internal validity
 - Omitted variable bias
 - Wrong functional form
 - Errors-in-variables
 - Sample selection
 - Simultaneous causality
- Threats to external validity
- RCTs

OLS and estimators (before midterm)

Check if you know the following

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Know the following:

- 3 OLS Assumptions (Key Concept 4.3)
- Formulas for $\hat{\beta}_1$ and $\hat{\beta}_0$ (Key Concept 4.2)
- Regression Output : ESS, TSS, SER, RMSE, R^2 , $\hat{Y}_i$, $\hat{u}_i$, $SE(\hat{\beta}_1)$
- Hypothesis testing on β_1 (two-sided and one-sided)
- Heteroskedasticity and Homoskedasticity
- What extra assumption gives you the Gauss-Markov Theorem? : Homoskedasticity!

Causal treatment effect (before midterm)

Check if you know the following

$$ATE = E[Y_i^1 - Y_i^0]$$

$$ATT = E[Y_i^1 - Y_i^0 | D_i = 1]$$

Know the following:

- How to compute ATE and ATT
- How to compute observed difference and selection bias
- Also could think of sign of selection bias
 - For example, aspirin: Individuals who chose to take aspirin would have higher headache than others who didn't choose aspirin if both of two groups didn't take aspirin
 - Selection bias : $E[Y_i^0 | D_i = 1] - E[Y_i^0 | D_i = 0]$ is positive by logic above (Here outcome is level of headache)

Multiple Regression Model

How to interpret coefficients

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

How to interpret coefficients:

- β_0 is population intercept, expected value of Y_i when all X s are 0
- Other β 's - change in Y_i for a 1 unit change in the associated X_i (holding all other X 's constant)
 - In math terms - $\beta_1 = \Delta Y / \Delta X_1$, **holding** $X_2, X_3, \dots, X_k$ **constant**
- If X is a dummy variable, coefficient (β) is the difference in expected Y between the category ($X = 1$) and the omitted group ($X = 0$)

Multiple Regression Model

Assumptions

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

Least Square Assumptions for Multiple Regression:

1. $E(u|X_1 = x_1, \dots, X_k = x_k) = 0$
2. $(X_{1i}, \dots, X_{ki}, Y_i)$, $i = 1, 2, \dots, n$, are i.i.d.
3. Large outliers are rare
 - Or, $E(X_{1i}^4) < \infty, \dots, E(X_{ki}^4) < \infty, E(Y_i^4) < \infty$.
4. There is no perfect multicollinearity ($\leftarrow$ New!)

Multiple Regression Model

Multicollinearity

$X_1, X_2, \dots, X_k$ are *perfectly multicollinear* if we can find numbers $a_1, a_2, \dots, a_k$ such that

$$X_1 = a_1 + a_2 X_2 + \dots + a_k X_k$$

Multiple Regression Model

Multicollinearity

Example: Age Ranges

- $Young = 1$ if $Age < 25$, and $Young = 0$ otherwise
- $MiddleAge = 1$ if $25 \leq Age < 55$, and $MiddleAge = 0$ otherwise
- $Old = 1$ if $55 < Age$, and $Old = 0$ otherwise

$Young$, $MiddleAge$, Old are mutually exclusive and exhaustive, which means you must be one of three groups. Hence we have the following relation:

$$Young = 1 - 1 \times MiddleAge - 1 \times Old$$

Check if this works by plugging in possible values for $MiddleAge$ and Old

Multiple Regression Model

Multicollinearity - Dummy Variable Trap

What is wrong with the following regression?

$$Y_i = \beta_0 + \beta_1 \text{Young}_i + \beta_2 \text{MiddleAge}_i + \beta_3 \text{Old}_i + u_i$$

Due to perfect multicollinearity of *Young*, *MiddleAge*, *Old*, so violation of assumption (4)

Multiple Regression Model

Multicollinearity - Dummy Variable Trap

Solution: Exclude one of the categories. For example,

$$Y_i = \beta_0 + \beta_1 \text{Young}_i + \beta_2 \text{MiddleAge}_i + u_i$$

- *Old* is the excluded category (reference group)
- β_1 is effect of *Young* relative to excluded category *Old*
- β_2 is effect of *MiddleAge* relative to excluded category *Old*
- β_0 is effect of excluded category *Old*
- Interpretation and value of coefficients changes depending on which category you exclude

Multiple Regression Model

Multicollinearity - Imperfect Multicollinearity

Imperfect multicollinearity:

- Occurs when regressors are highly correlated
- Does *not* violate assumption (4) or imply OVB (OVB : explained later)
- Increases the standard errors

Multiple Regression Model

Joint Hypothesis Testing : Set up null and alternative hypotheses

Multiple Regression Model, again:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

Examples of Joint Null Hypothesis:

- $\beta_1 = 0, \beta_2 = 0, \dots, \beta_k = 0 \rightarrow q = k$
- $\beta_1 = 0, \beta_2 = 0 \rightarrow q = 2$
- $\beta_1 = \beta_2 \rightarrow q = 1$

q = number of restrictions (count equal signs)

Should know how to set up null and alternative hypotheses

- Example : $H_0 : \beta_1 = 0, \beta_2 = 0$, H_1 : either $\beta_1 \neq 0$ or $\beta_2 \neq 0$ or both

Multiple Regression Model

Joint Hypothesis Testing : Example in large sample

Example: $\text{birthweight}_i = \beta_0 + \beta_1 \text{Smoker}_i + \beta_2 \text{alcohol}_i + \beta_3 \text{nprevist}_i + u$

```
. reg birthweight smoker alcohol nprevist, robust
```

Linear regression	Number of obs	=	3,000
	F(3, 2996)	=	59.48
	Prob > F	=	0.0000
	R-squared	=	0.0729
	Root MSE	=	570.47

birthweight	Robust		t	P> t	[95% Conf. Interval]	
	Coef.	Std. Err.				
smoker	-217.5801	26.10764	-8.33	0.000	-268.7708	-166.3894
alcohol	-30.49129	72.59671	-0.42	0.675	-172.8357	111.8531
nprevist	34.06991	3.608326	9.44	0.000	26.99487	41.14496
_cons	3051.249	43.71445	69.80	0.000	2965.535	3136.962

```
. test alcohol nprevist
```

```
( 1) alcohol = 0
```

```
( 2) nprevist = 0
```

```
F( 2, 2996) = 44.71
```

- $H_0 : \beta_2 = 0, \beta_3 = 0$
- $H_1 : \beta_2 \neq 0$, or $\beta_3 \neq 0$, or both ($q = 2$)
- F-statistic is 44.71, larger than 5% critical value of $\chi^2_{2/2, 3.00}$.
- Reject the null that *alcohol* and *nprevist* don't have effect on *birthweight*.

Multiple Regression Model

Joint Hypothesis Testing : F-statistic

You might be asked to deal with the F-statistic when there are only two restrictions ($q = 2$). Then:

$$F = \frac{1}{2} \left(\frac{t_1^2 - t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right)$$

where:

- t_i is the t-statistic for the i th hypothesis
- $\hat{\rho}_{t_1, t_2}$ is the correlation between the two t-statistics

Threats to Internal Validity

- Five sources of threats to internal validity
 - **Omitted variable bias**
 - Wrong functional form
 - Errors-in-variables
 - Sample selection
 - Simultaneous causality

Threats to Internal Validity

Omitted Variable Bias : Conditions for OVB

Two conditions for OVB to occur:

1. The omitted variable is correlated with the included regressor
2. The omitted variable is a determinant of the dependent variable

In mathematical form, consider a regression with omitted variable Z ,

$$Y_i = \beta_0 + \beta_1 X_i + u_i = \beta_0 + \beta_1 X_i + (\gamma Z_i + v_i)$$

Then two conditions are as follows,

1. $\text{Cov}(X, Z) \neq 0$
2. $\gamma \neq 0$

Which OLS assumption is violated if we have OVB?

Assumption 1: $E[u_i | X_i = x_i] = 0$

Threats to Internal Validity

Omitted Variable Bias : Sign of OVB

The omitted variable bias formula:

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \gamma \frac{\text{Cov}(X, Z)}{\text{Var}(X)}$$

Hence, sign of OVB is determined by $\text{Cov}(X, Z)$ and γ .

Threats to Internal Validity

Omitted Variable Bias : Example of OVB

Example: Effect of being in preschool on earning in adulthood

$$Earning_i = \beta_0 + \beta_1 Preschool_i + u_i$$

Here *Preschool* is 1 if an individual *i* was in preschool in childhood, and 0 otherwise. Does *Preschool* suffer by omitted variable bias problem?

Omitted variable : Parents' Income

1. $Cov(Preschool, Parents' Income) > 0$ because Preschool is expensive
2. $\gamma > 0$, Parents' income might have a positive impact on earning

Sign of OVB : $Sign(\gamma) \times Sign(Cov(X, Z)) = (+) \times (+) = (+)$

- This means $\hat{\beta}_1 > \beta_1$
- $\hat{\beta}_1$ will overstate the change in earning associated with going to preschool

Threats to Internal Validity

Omitted Variable Bias : How to solve OVB

How to solve omitted variable bias problem?

- Randomized Controlled Experiment (usually infeasible)
 - In the previous example, make treatment (being in a preschool) being randomly assigned to each individual, so that there is no correlation between being in a preschool and parents' income
 - This way could make $Cov(X, Z) = 0$, even though γ is still nonzero
- **Include the omitted variable in the regression**

Threats to Internal Validity

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Threats to Internal Validity

Wrong functional form : Nonlinearity in regression

What if the relationship between X and Y is NOT linear?

- Then linear regression $Y_i = \beta_0 + \beta_1 X_i + u_i$ leads to biased estimator of effect on Y of X

How to solve this problem? : use a nonlinear function in regression

- Log transforms of Y and/or X
- Polynomials in X: for example, quadratic formula is as follows:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i$$

- Interactions between regressors

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 (X_i \times Z_i) + u_i$$

Threats to Internal Validity

Wrong functional form : Interpret coefficient with log form

Variable with the log will have the percent change

- Log-linear : $\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i$
 - 1 unit change in X is associated with $100 \times \beta_1\%$ change in Y
- Linear-log : $Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i$
 - 1% unit change in X is associated with $0.01 \times \beta_1$ change in Y
- Log-log : $\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i$
 - 1% unit change in X is associated with $\beta_1\%$ change in Y

How to interpret change in variable in log form?

- $\ln(X)$ increases by 0.01 $\approx$ X increases by 1%
- For example, if $\ln(X)$ increases by 0.025, then X goes up by 2.5%

Threats to Internal Validity

Wrong functional form : Polynomials in regressions

We could have polynomials in X as follows

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \cdots + \beta_r X_i^r + u_i$$

- Estimation, hypothesis testing, etc. proceeds as in the multiple regression model using OLS
- the coefficients are difficult to interpret: can't change X holding X^2 constant
- Test null hypothesis that Y is linear in X :
 - $H_0 : \beta_2 = \beta_3 = \cdots = \beta_r = 0$
 - H_1 : At least one of β is nonzero
 - This is a joint hypothesis test: use F-statistic with $q = r - 1$

Threats to Internal Validity

Wrong functional form : Why use Interaction Terms

Example: Earnings, Age, Preschool $Pre_i = 1$ if person i went to preschool

- $Earn_i = \beta_0 + \beta_1 Pre_i + \beta_2 Age_i + u_i$ assumes effect of Age on $Earning$ (β_2) is the same for preschool and non-preschool
- How to make slopes different across Pre ? Include interaction term, $Age_i \times Pre_i$

Threats to Internal Validity

Wrong functional form : Regression with Interaction Term

Regression with interaction term of continuous variable and binary variable

$$Earn_i = \beta_0 + \beta_1 Pre_i + \beta_2 Age_i + \beta_3 Pre_i \times Age_i + u_i$$

Consider two cases:

- $Preschool = 0$: $\widehat{Earn} = \hat{\beta}_0 + \hat{\beta}_2 Age$
- $Preschool = 1$:

$$\widehat{Earn} = \hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 Age + \hat{\beta}_3 Age = \underbrace{(\hat{\beta}_0 + \hat{\beta}_1)}_{\text{intercept}} + \underbrace{(\hat{\beta}_2 + \hat{\beta}_3)}_{\text{slope}} Age$$

- Estimates a regression line for each group of Pre
- β_1 is the incremental intercept, when Pre is changed from 0 to 1
- β_3 is the incremental effect of Age , when Pre is changed from 0 to 1

Threats to Internal Validity

Wrong functional form : various null hypotheses with interaction terms

$$Earn_i = \beta_0 + \beta_1 Pre_i + \beta_2 Age_i + \beta_3 Pre_i \times Age_i + u_i$$

Can test various hypotheses:

- Test that effect of *Age* is the same for *Preschool* and *Non-Preschool*?
 - which means slope is same across *Pre*, so $H_0 : \beta_3 = 0$
- Test that the regression lines have the same intercept between *Preschool* and *Non-Preschool*?
 - $H_0 : \beta_1 = 0$
- Test that the regression lines are the same?
 - which means slope and intercept are same across *Pre*, so $H_0 : \beta_1 = 0, \beta_3 = 0$
 - Need joint hypotheses testing so use an F-statistic (Here, $q = 2$.)

Threats to Internal Validity

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 - Omitted variable bias
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 - **Errors-in-variables**
 - Sample selection
 - Simultaneous causality

Threats to Internal Validity

Errors in Variables : error in X or Y

How can it arise?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

What is effect on estimators if ...

- X is measured with error?
 - $\hat{\beta}_1$ is biased and inconsistent! Specifically biased *towards zero*.
 - Intuition: In extreme case, X is all noise $\rightarrow$ no effect of X on Y, which means $\beta_1 = 0$
- Y is measured with error?
 - $\hat{\beta}_1$ is still unbiased and consistent (assumed error uncorrelated with X)
- Check Exercise 9.2 in the textbook

How to solve this problem? : Get a better data if possible, develop a specific model of errors, or instrumental variable regression

Threats to Internal Validity

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Threats to Internal Validity

Sample selection : When this is a problem

Sample selection bias arises when a selection process:

1. influences the availability of data and
2. that process is related to the dependent variable

Example: Effect of a 6 month Diet program. 1000 people, half in a diet program and half not. Interested in the effect on Weight Loss of the Diet program, so you set up the regression as follows.

$$WeightLoss_i = \beta_0 + \beta_1 Diet_i + u_i$$

You sample only people who *finish* Diet program, as well as those not in the diet program: What's the problem of this sampling process?

Threats to Internal Validity

Sample selection : How to solve this problem

You sample only people who *finish* Diet program, as well as those not in the diet program: What's the problem of this sampling process?

- A person might prematurely leave the Diet if it isn't working, so being in the program means the diet worked well for you
- Lost the people for whom the diet didn't work (availability of data)
- Have people who succeed to lose weight by program only (related to dependent variable)
- Hence $\hat{\beta}_1$ will overstate the effect of the program

How to solve this problem? : Collect the sample in a different way, RCT, or develop a model of sample selection and estimate this model

Threats to Internal Validity

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Threats to Internal Validity

Simultaneous causality : How to solve this problem

I want to know X 's causal effect on Y , *but* Y also affects X

Example: Effect of smoking (X) on mental health (Y)

- Smoking (X) would affect one's mental health (Y)
- If a person is stressed (Y), however, likely to smoke more (X)
- In this case, $\hat{\beta}$ is biased and inconsistent

How to solve this problem? : RCT, or instrumental variable regression

Threats to External Validity

Two sources of threats

Can we apply the regression results obtained from one context to another?

Difference in population

- Is your sample close enough to another group of population? or similar to general population?
- i.e. Animal and human, people in different states, college graduates and general people, etc.

Difference in Setting

- Is there any difference in settings? i.e. institutional environment, laws, or physical environment
- i.e. public or religious universities, price change in tobacco, etc.

RCT (Randomized Controlled Trials)

Definition of RCT

RCT, or randomized experiments

- Random: Subjects *randomly* assigned to treatment and control groups
- Controlled: There is a *control* group which doesn't receive treatment
- Trial: Treatment is assigned with subjects having no choice

In RCT, treatment, D_i is *independent* of Y_i^1 and Y_i^0 . Hence,

$$E[Y_i^1 | D_i = 1] = E[Y_i^1 | D_i = 0] = E[Y_i^1]$$

$$E[Y_i^0 | D_i = 1] = E[Y_i^0 | D_i = 0] = E[Y_i^0]$$

RCT (Randomized Controlled Trials)

Why RCT is useful

In RCT, ATE and ATT are equal to observed difference

$$\underbrace{E[Y_i|D_i = 1] - E[Y_i|D_i = 0]}_{\text{Observed difference}} = E[Y_i^1|D_i = 1] - E[Y_i^0|D_i = 0]$$
$$= \underbrace{E[Y_i^1] - E[Y_i^0]}_{\text{ATE}}$$

$$\underbrace{E[Y_i|D_i = 1] - E[Y_i|D_i = 0]}_{\text{Observed difference}} = E[Y_i^1|D_i = 1] - E[Y_i^0|D_i = 1]$$
$$= \underbrace{E[Y_i^1|D_i = 1] - E[Y_i^0|D_i = 1]}_{\text{ATT}}$$
$$+ \underbrace{E[Y_i^0|D_i = 1] - E[Y_i^0|D_i = 0]}_{\text{Selection bias (=0)}}$$